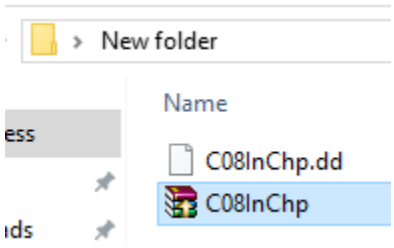
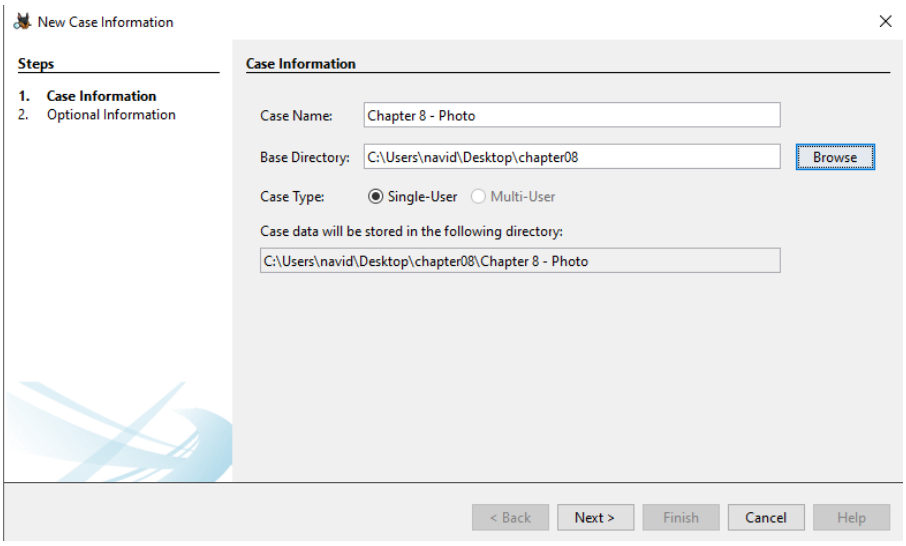
Navid Baghaei

Lab 8

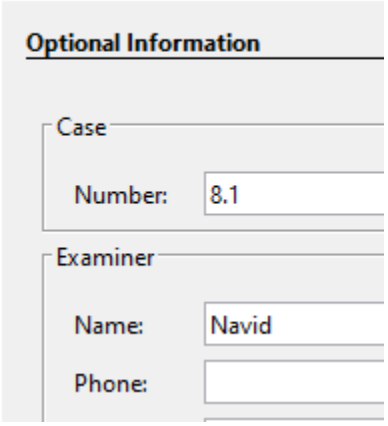
TASK 1

1. Started my windows VM and copied the C08InChp file and decompressed it.

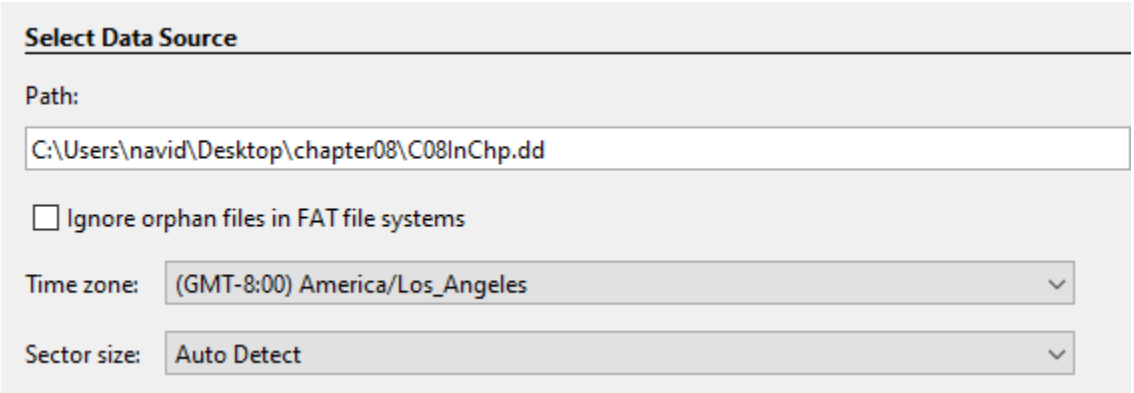




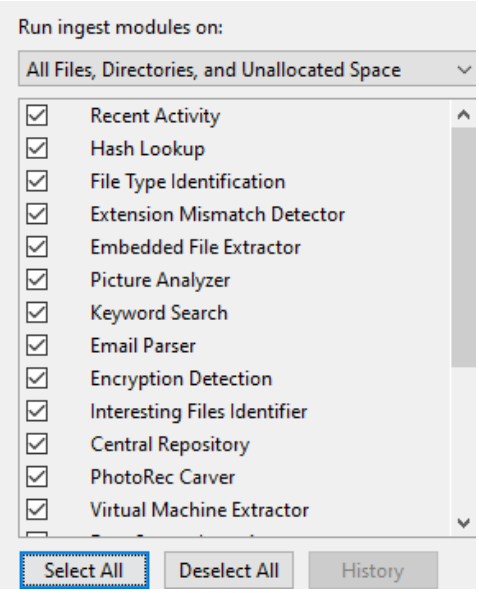
Made a new case named Chapter 8 – Photo



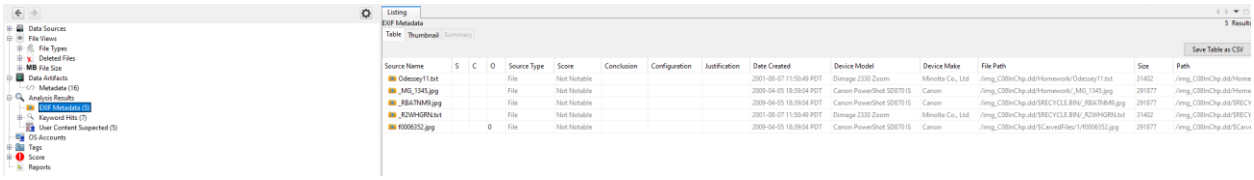
Assigned myself as the examiner and gave the case a number.



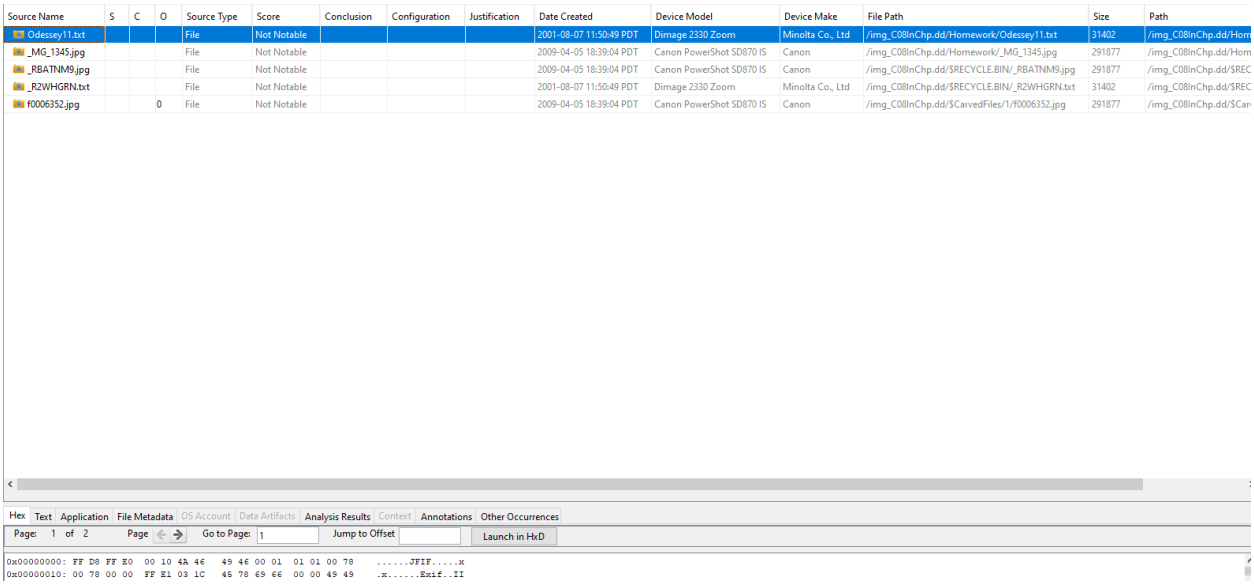
Added the provided image file as the path.



Selected all Ingest Modules, then pressed next and finish.

2. 

Selected EXIF Metadata under Analysis Results on the left tab.



Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Date Created	Device Model	Device Make	File Path	Size	Path
Odessey11.txt				File	Not Notable				2001-08-07 11:50:49 PDT	Dimage 2330 Zoom	Minolta Co., Ltd	/img_C08lnChp.dd/Homework/Odessey11.txt	31402	/img_C08lnChp.dd/Hom
_MG_1345.jpg				File	Not Notable				2009-04-05 18:39:04 PDT	Canon PowerShot SD870 IS	Canon	/img_C08lnChp.dd/Homework/_MG_1345.jpg	291877	/img_C08lnChp.dd/Hom
_RBATNM9.jpg				File	Not Notable				2009-04-05 18:39:04 PDT	Canon PowerShot SD870 IS	Canon	/img_C08lnChp.dd/$RECYCLE.BIN/_RBATNM9.jpg	291877	/img_C08lnChp.dd/$REC
_R2WHGRN.txt				File	Not Notable				2001-08-07 11:50:49 PDT	Dimage 2330 Zoom	Minolta Co., Ltd	/img_C08lnChp.dd/$RECYCLE.BIN/_R2WHGRN.txt	31402	/img_C08lnChp.dd/$REC
f0006352.jpg			0	File	Not Notable				2009-04-05 18:39:04 PDT	Canon PowerShot SD870 IS	Canon	/img_C08lnChp.dd/$CarvedFiles/1/f0006352.jpg	291877	/img_C08lnChp.dd/$Car

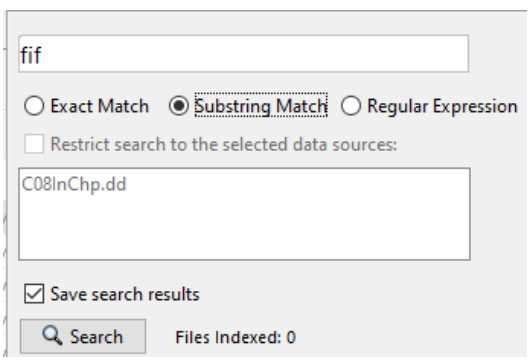
Hex | Text | Application | File Metadata | OS Account | Data Artifacts | Analysis Results | Context | Annotations | Other Occurrences

Page: 1 of 2 Page Go to Page: 1 Jump to Offset Launch in HxD

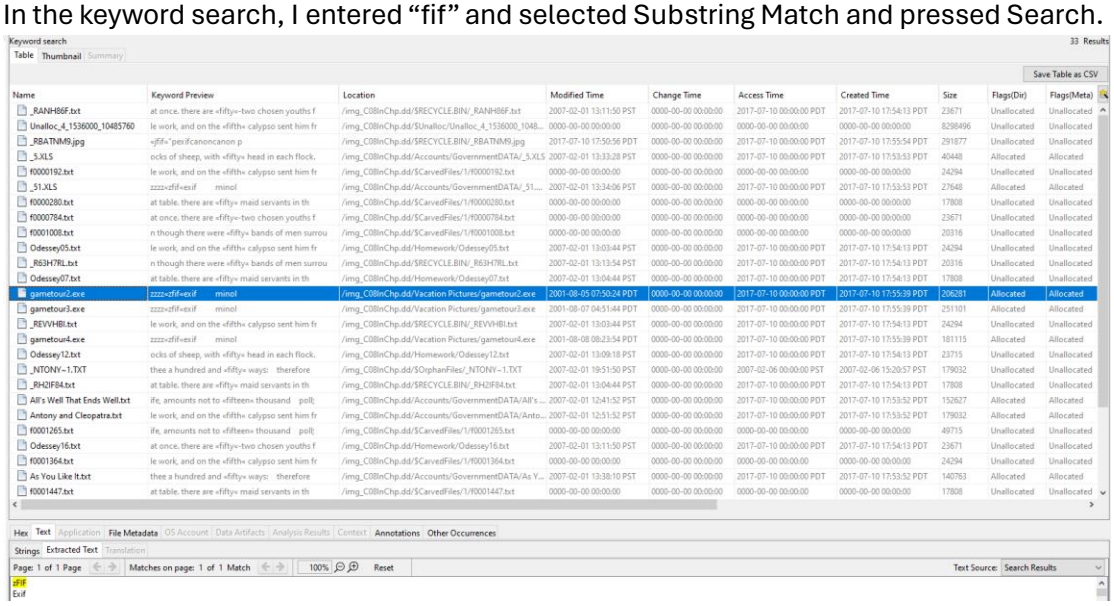
```
0x00000000: FF D8 FF E0  00 10 4A 46   49 46 00 01  01 01 00 78    ......JFIF.....x
0x00000010: 00 78 00 00  FF E1 03 1C   45 78 69 66  00 00 49 49    .x......Exif..II
```

Selecting Odessey11.txt and viewing the hex tab, we can see the file name extension is JFIF instead of txt.

3.



In the keyword search, I entered “fif” and selected Substring Match and pressed Search.



Keyword search — 33 Results

Table | Thumbnail | Summary

Save Table as CSV

Name	Keyword Preview	Location	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)
_RANH86F.txt	at once. there are «fifty»-two chosen youths f	/img_C08lnChp.dd/$RECYCLE.BIN/_RANH86F.txt	2007-02-01 13:11:50 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:54:13 PDT	23671	Unallocated	Unallocated
Unalloc_4_1536000_10485760	le work, and on the «fifth» calypso sent him fr	/img_C08lnChp.dd/$Unalloc/Unalloc_4_1536000_1048...	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	8298496	Unallocated	Unallocated
_RBATNM9.jpg	«jfif»"pexifcanoncanon p	/img_C08lnChp.dd/$RECYCLE.BIN/_RBATNM9.jpg	2017-07-10 17:50:56 PDT	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:55:54 PDT	291877	Unallocated	Unallocated
_5.XLS	ocks of sheep, with «fifty» head in each flock.	/img_C08lnChp.dd/Accounts/GovernmentDATA/_5.XLS	2007-02-01 13:33:28 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:53:53 PDT	40448	Allocated	Allocated
f0000192.txt	le work, and on the «fifth» calypso sent him fr	/img_C08lnChp.dd/$CarvedFiles/1/f0000192.txt	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	24294	Unallocated	Unallocated
_51.XLS	zzzz«zfif«exif minol	/img_C08lnChp.dd/Accounts/GovernmentDATA/_51....	2007-02-01 13:34:06 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:53:53 PDT	27648	Allocated	Allocated
f0000280.txt	at table. there are «fifty» maid servants in th	/img_C08lnChp.dd/$CarvedFiles/1/f0000280.txt	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	17808	Unallocated	Unallocated
f0000784.txt	at once. there are «fifty»-two chosen youths f	/img_C08lnChp.dd/$CarvedFiles/1/f0000784.txt	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	23671	Unallocated	Unallocated
f0001008.txt	n though there were «fifty» bands of men surrou	/img_C08lnChp.dd/$CarvedFiles/1/f0001008.txt	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	20316	Unallocated	Unallocated
Odessey05.txt	le work, and on the «fifth» calypso sent him fr	/img_C08lnChp.dd/Homework/Odessey05.txt	2007-02-01 13:03:44 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:54:13 PDT	24294	Unallocated	Unallocated
_R63H7RL.txt	n though there were «fifty» bands of men surrou	/img_C08lnChp.dd/$RECYCLE.BIN/_R63H7RL.txt	2007-02-01 13:13:54 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:54:13 PDT	20316	Unallocated	Unallocated
Odessey07.txt	at table. there are «fifty» maid servants in th	/img_C08lnChp.dd/Homework/Odessey07.txt	2007-02-01 13:04:44 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:54:13 PDT	17808	Unallocated	Unallocated
gametour2.exe	zzzz«zfif«exif minol	/img_C08lnChp.dd/Vacation Pictures/gametour2.exe	2001-08-05 07:50:24 PDT	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:55:39 PDT	206281	Allocated	Allocated
gametour3.exe	zzzz«zfif«exif minol	/img_C08lnChp.dd/Vacation Pictures/gametour3.exe	2001-08-07 04:51:44 PDT	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:55:39 PDT	251101	Allocated	Allocated
_REVVHBI.txt	le work, and on the «fifth» calypso sent him fr	/img_C08lnChp.dd/$RECYCLE.BIN/_REVVHBI.txt	2007-02-01 13:03:44 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:54:13 PDT	24294	Unallocated	Unallocated
gametour4.exe	zzzz«zfif«exif minol	/img_C08lnChp.dd/Vacation Pictures/gametour4.exe	2001-08-08 08:23:54 PDT	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:55:39 PDT	181115	Allocated	Allocated
Odessey12.txt	ocks of sheep, with «fifty» head in each flock.	/img_C08lnChp.dd/Homework/Odessey12.txt	2007-02-01 13:09:18 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:54:13 PDT	23715	Unallocated	Unallocated
_NTONY~1.TXT	thee a hundred and «fifty» ways: therefore	/img_C08lnChp.dd/$OrphanFiles/_NTONY~1.TXT	2007-02-01 19:51:50 PST	0000-00-00 00:00:00	2007-02-06 00:00:00 PST	2007-02-06 15:20:57 PST	179032	Unallocated	Unallocated
_RH2IF84.txt	at table. there are «fifty» maid servants in th	/img_C08lnChp.dd/$RECYCLE.BIN/_RH2IF84.txt	2007-02-01 13:04:44 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:54:13 PDT	17808	Unallocated	Unallocated
All's Well That Ends Well.txt	ife, amounts not to «fifteen» thousand poll;	/img_C08lnChp.dd/Accounts/GovernmentDATA/All's ...	2007-02-01 12:41:52 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:53:52 PDT	152627	Allocated	Allocated
Antony and Cleopatra.txt	le work, and on the «fifth» calypso sent him fr	/img_C08lnChp.dd/Accounts/GovernmentDATA/Anto...	2007-02-01 12:51:52 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:53:52 PDT	179032	Allocated	Allocated
f0001265.txt	ife, amounts not to «fifteen» thousand poll;	/img_C08lnChp.dd/$CarvedFiles/1/f0001265.txt	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	49715	Unallocated	Unallocated
Odessey16.txt	at once. there are «fifty»-two chosen youths f	/img_C08lnChp.dd/Homework/Odessey16.txt	2007-02-01 13:11:50 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:54:13 PDT	23671	Unallocated	Unallocated
f0001364.txt	le work, and on the «fifth» calypso sent him fr	/img_C08lnChp.dd/$CarvedFiles/1/f0001364.txt	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	24294	Unallocated	Unallocated
As You Like It.txt	thee a hundred and «fifty» ways: therefore	/img_C08lnChp.dd/Accounts/GovernmentDATA/As Y...	2007-02-01 13:38:10 PST	0000-00-00 00:00:00	2017-07-10 00:00:00 PDT	2017-07-10 17:53:52 PDT	140763	Allocated	Allocated
f0001447.txt	at table. there are «fifty» maid servants in th	/img_C08lnChp.dd/$CarvedFiles/1/f0001447.txt	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	17808	Unallocated	Unallocated

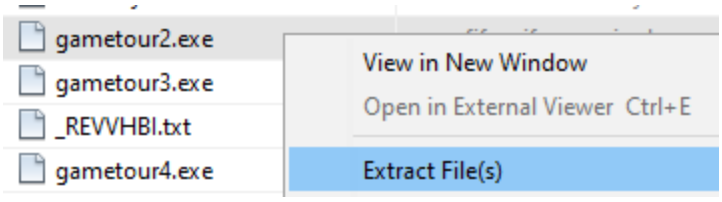
Hex | Text | Application | File Metadata | OS Account | Data Artifacts | Analysis Results | Context | Annotations | Other Occurrences

Strings | Extracted Text | Translation

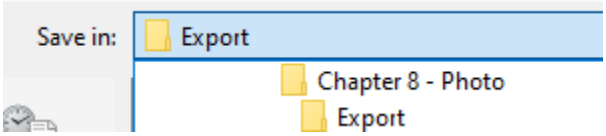
Page: 1 of 1 Page Matches on page: 1 of 1 Match 100% Reset Text Source: Search Results

zFIF
Exif

A page of results showed up. Selecting “gametour2.exe” shows that the hex file type is “zFIF” which is what we searched for.

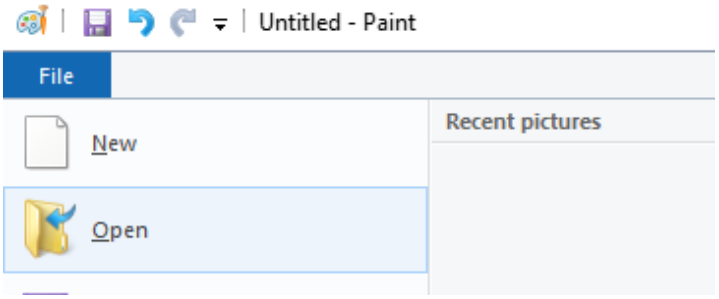


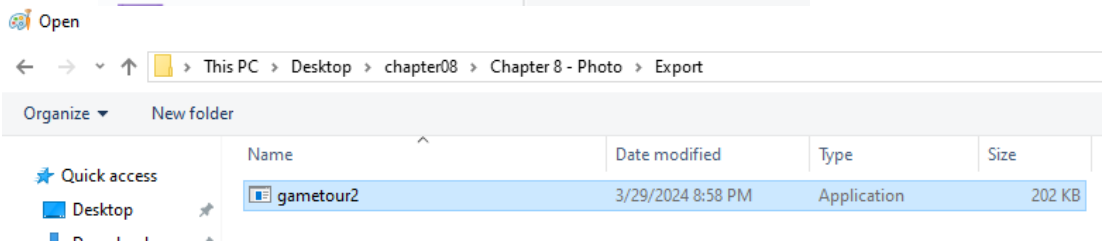
Extracted the gametour2.exe file,



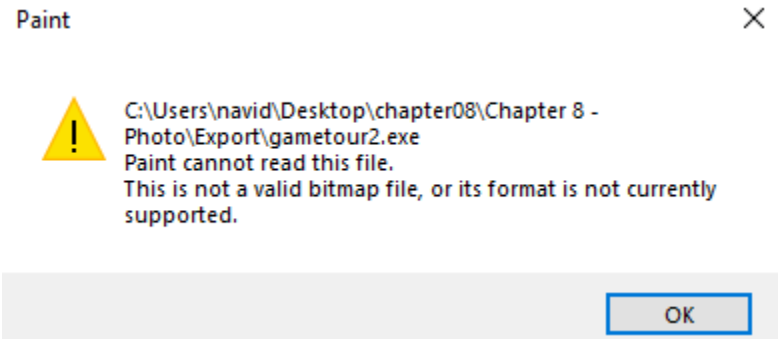
And saved it in the case folder's export folder.

4.

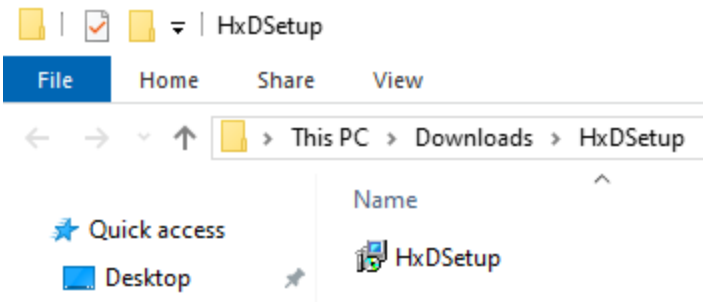




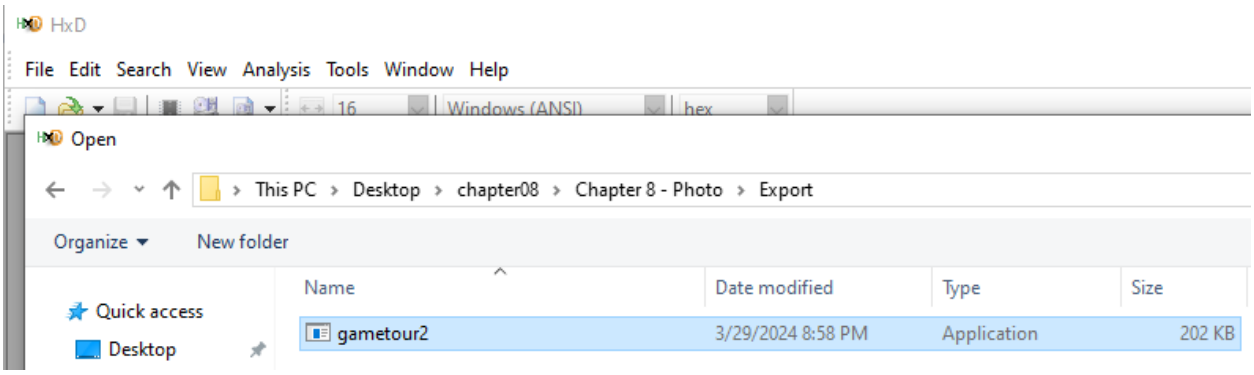
Launched Microsoft Paint, selected open under file, and selected the gametour2 application.



The file cannot be opened because paint doesn't support .exe.



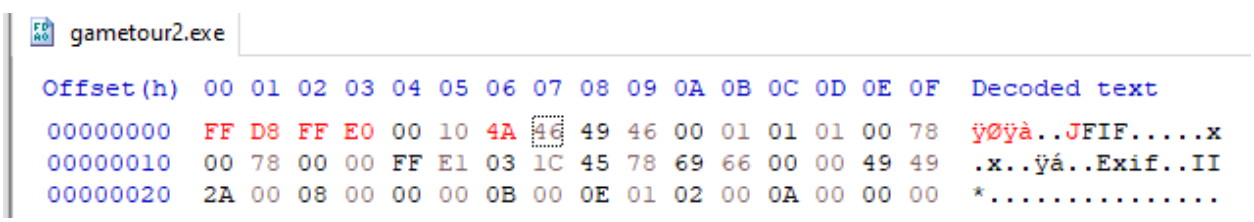
Headed to mh-nexus.de and downloaded the English version. Extracted it and ran the installation.



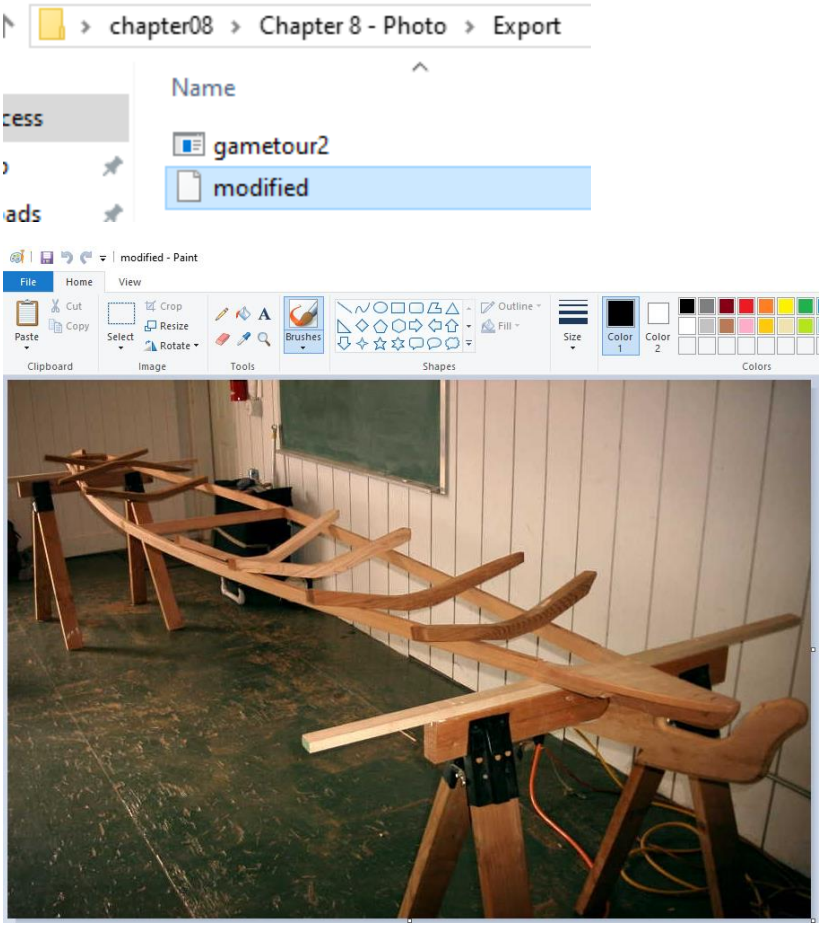
Once inside the program, I went to file, and opened the gametour2.exe file.

```
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F  Decoded text
00000000  7A 7A 7A 7A 00 10 7A 46 49 46 00 01 01 01 00 78  zzzz..zFIF.....x
00000010  00 78 00 00 FF E1 03 1C 45 78 69 66 00 00 49 49  .x..ÿá..Exif..II
00000020  2A 00 08 00 00 00 0B 00 0E 01 02 00 0A 00 00 00  *...............
00000030  92 00 00 00 0F 01 02 00 12 00 00 00 9C 00 00 00  '...........œ...
00000040  10 01 02 00 12 00 00 00 AE 00 00 00 12 01 03 00  ........®.......
00000050  01 00 00 00 01 00 08 00 1A 01 05 00 01 00 00 00  ................
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F  Decoded text
00000000  FF D8 FF E0 00 10 7A 46 49 46 00 01 01 01 00 78  ÿØÿà..zFIF.....x
00000010  00 78 00 00 FF E1 03 1C 45 78 69 66 00 00 49 49  .x..ÿá..Exif..II
00000020  2A 00 08 00 00 00 0B 00 0E 01 02 00 0A 00 00 00  *
```

Changed the first 4 bytes to the JFIF header values. The changes appear in red.

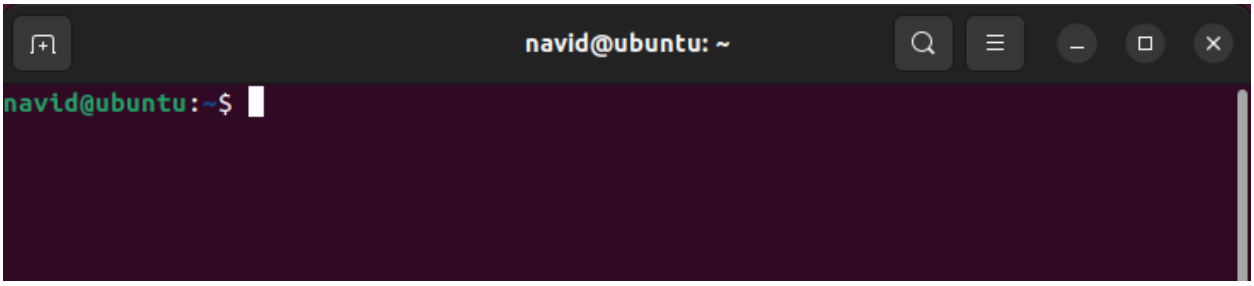


Replaced the z in “zFIF” with the letter J. The change is also reflected on the 6th byte as 7A changes to 4A.



Saved the file as “modified.jpeg”. Opening it in paint now shows the image!

TASK 2

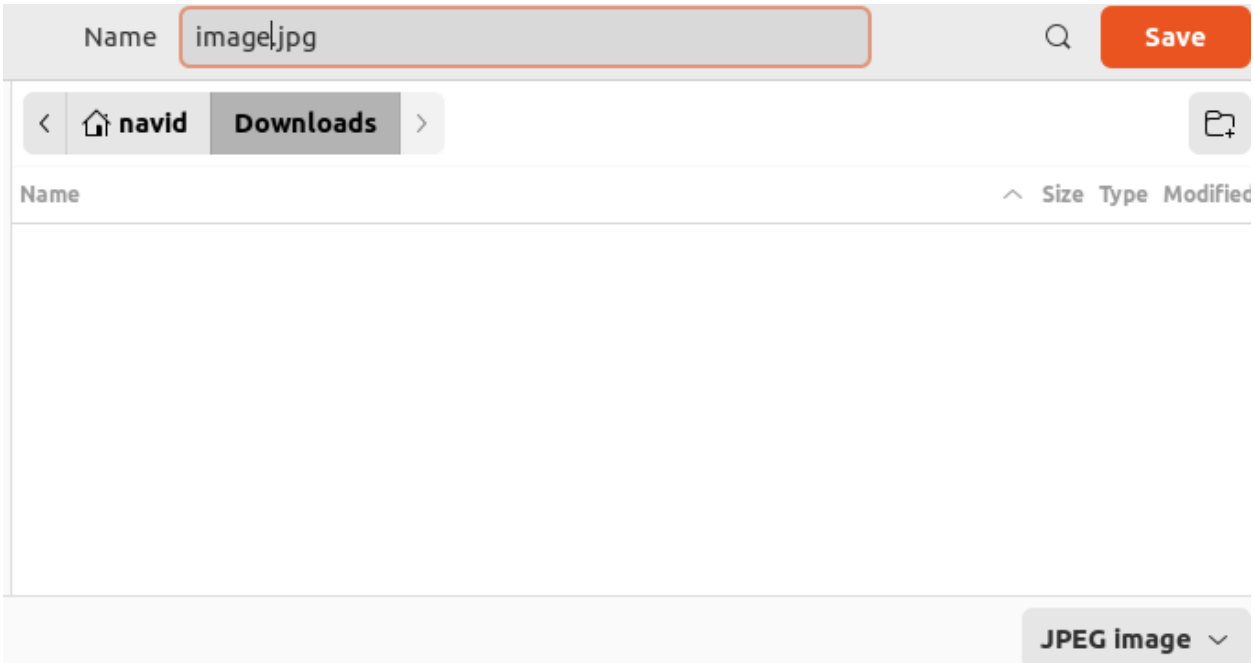
1. 

Launched Ubuntu VM and opened the terminal.

```
navid@ubuntu: ~
navid@ubuntu:~$ sudo apt update -y
[sudo] password for navid:
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [110 kB]
Hit:2 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease [119 kB]
Get:4 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease [109 kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [436 k
B]
```

```
navid@ubuntu:~$ sudo apt install steghide -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libmcrypt4
Suggested packages:
  libmcrypt-dev mcrypt
The following NEW packages will be installed:
  libmcrypt4 steghide
0 upgraded, 2 newly installed, 0 to remove and 155 not upgraded.
```

Downloaded and installed steghide.

2. 

Downloaded an image from google and saved it as “image.jpg”.

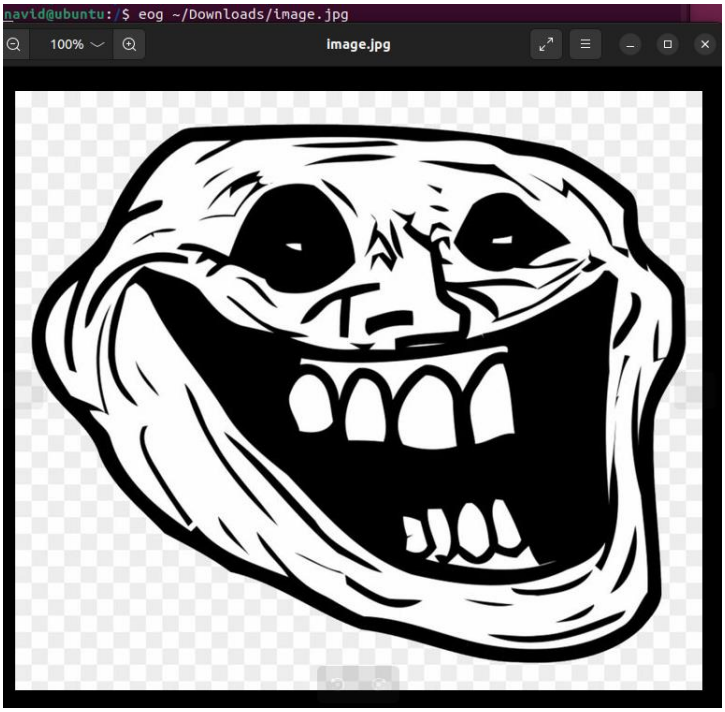
3.
```
navid@ubuntu:~$ echo "Launch Codes: 123123" > secret.txt
navid@ubuntu:~$
```

Created a secret message that will be hidden in the JPG image.

4.

```
navid@ubuntu:~$ steghide embed -ef secret.txt -cf ~/Downloads/image.jpg
Enter passphrase:
Re-Enter passphrase:
embedding "secret.txt" in "/home/navid/Downloads/image.jpg"... done
```

Used steghide to hide the secret message within the jpg image.



There is no visual difference in the image even after embedding the secret message in it.

5.

```
navid@ubuntu: ~/Downloads
navid@ubuntu:~$ cd ~/Downloads/
navid@ubuntu:~/Downloads$ steghide extract -sf image.jpg
Enter passphrase:
wrote extracted data to "secret.txt".
navid@ubuntu:~/Downloads$ cat secret.txt
Launch Codes: 123123
```

Navigated to where the image file was, and extracted and displayed the secret file from the image.